AdaptLearn

Agentic AI for Inclusive Education

An autonomous multi-agent system that personalizes learning for students with dyslexia, ADHD, and other learning disabilities

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1. Problem Description and Context

1.1. Overview

Education systems worldwide operate on a single implicit assumption: all students learn the same way. Textbooks are written at a fixed reading level. Lessons run for fixed durations. Assessments use standard formats. For the majority of students, this is merely inconvenient. For the approximately **one in five students** who live with a learning disability, it is a fundamental barrier that shapes their entire academic trajectory.

Learning disabilities — including dyslexia, ADHD, dyscalculia, and dysgraphia — are neurological conditions that affect how the brain processes information. They are not indicators of intelligence. They are not temporary. Yet the current educational infrastructure, even in well-resourced countries, is systematically unable to serve this population at scale.

AdaptLearn is our answer to this challenge: an autonomous multi-agent AI system that perceives each student's unique cognitive profile, adapts educational content in real time, monitors engagement, and generates actionable reports for educators — all without requiring continuous manual intervention.

1.2. The Scale of the Problem: Key Statistics

> **Why this problem demands an AI-scale solution:** The data below makes it clear that human-only interventions, however well-intentioned, cannot operate at the scale this crisis demands.

1.2.1. Prevalence of Learning Disabilities

- **1 in 5 students** — approximately 20% of the global student population — has a learning disability of some kind. This figure, established by the *National Center for Learning Disabilities (NCLD, 2017)*, makes learning disabilities the most common category of educational need worldwide.

- **Dyslexia** is the most prevalent form, affecting **15–20%** of the population and accounting for **80–90%** of all learning disability diagnoses (*Yale Center for Dyslexia & Creativity, 2020*).

- **ADHD** affects **5–7%** of school-age children globally, according to the *World Health Organization (WHO, 2019)*, with significant overlap with other learning disabilities.

- **Dyscalculia** — a specific difficulty with numbers and mathematical reasoning — affects approximately **3–7%** of the population (*Butterworth et al., Trends in Cognitive Sciences, 2011*).

1.2.2. The Gap Between Need and Support

- Only **17%** of students with dyslexia receive appropriate, evidence-based support during their schooling (*British Dyslexia Association, 2018*).

- An estimated **two-thirds** of students with learning disabilities are never formally identified (*NCLD, 2017*). They navigate school labeled as "lazy" or "not trying hard enough."

- Students with unaddressed learning disabilities are **3 times more likely** to drop out of secondary education than their peers (*National Longitudinal Transition Study, US Department of Education, 2011*).
- The average student-to-specialist ratio in schools is **1 specialist per 500+ students** in most OECD countries (*OECD Education at a Glance, 2022*), making personalized 1-on-1 intervention structurally impossible.

1.2.3. The Cost of Inaction

> **Economic impact:** Adults with unaddressed learning disabilities earn **35% less** on average than peers with similar cognitive ability who received appropriate support. Learning disabilities are not a ceiling — inadequate support is (*NCLD, 2017; Raskind et al., Annals of Dyslexia, 1999*).

1.2.4. The Moroccan and African Context

- In Morocco, special educational needs are addressed by the *Centre de Ressources pour la Scolarisation des Enfants en Situation de Handicap (CRSEESH)*, but access remains highly limited to urban centers.
- The *UNESCO Global Education Monitoring Report (2020)* notes that Sub-Saharan Africa and North Africa have the widest gap between prevalence of learning difficulties and availability of support infrastructure.
- Digital transformation initiatives such as Morocco's *Vision Stratégique 2030* explicitly identify inclusive education as a national priority — making AI-powered accessibility tools a direct policy fit.

1.3. Why Existing Solutions Fall Short

> **The gap AdaptLearn fills:** No existing solution combines real-time behavioral observation, autonomous content adaptation, multi-disability support, and teacher-facing reporting in a single coherent system. This is the space that agentic AI uniquely enables.

2. Functional Requirements

2.1. System Overview

AdaptLearn is structured around five core functional modules, each handled by a dedicated autonomous agent. Together they form a closed loop: observe the student, adapt the content, monitor engagement, respond to exercises, and report to educators.

2.2. FR-01: Student Onboarding and Profile Creation

- The system shall conduct a short conversational onboarding session with a new student.

Table 1: Comparison of current approaches to supporting students with learning disabilities

Approach	Limitations	What is missing
Human tutors	Expensive, geographically limited, not available 24/7	Scalability
Static IEP plans	Written once per year, cannot adapt to daily changes	Real-time adaptation
Accessibility tools (e.g., text-to-speech)	Passive tools, no reasoning or feedback loop	Autonomous decision-making
Traditional e-learning	One-size-fits-all content, no learner modelling	Personalization
Chatbot tutors	Rule-based, no long-term memory, no behavioral observation	Agentic autonomy

- The Profile Agent shall infer the student's disability profile (dyslexia, ADHD, dyscalculia, or combinations) from questionnaire responses and initial interaction patterns.
- The learner model shall store: disability type(s), preferred content modality (text/audio/visual), optimal session duration, language level, and grade level.
- The profile shall be updatable at any time based on subsequent session data.
- Students shall be able to optionally provide an existing diagnosis to accelerate profile setup.

2.3. FR-02: Autonomous Content Adaptation

- Teachers shall be able to upload raw educational content (PDF, Word, plain text).
- The Adaptation Agent shall autonomously transform uploaded content according to each student's profile without manual configuration per student.
- For **dyslexic** students: the system shall apply shorter sentence structures, syllable markers on polysyllabic words, dyslexia-friendly fonts (OpenDyslexic), increased line spacing (minimum 1.8), and color overlay options.

- For **ADHD** students: the system shall chunk content into timed micro-sessions (5–10 minutes), add visible progress indicators, and include structured break prompts.
- The system shall generate audio versions of all text content via a text-to-speech tool.
- Adapted content shall be delivered through a web interface accessible on desktop and mobile.

2.4. FR-03: Real-Time Engagement Monitoring

- The Monitor Agent shall passively track student behavioral signals during each session, including: scroll speed, time-on-paragraph, response latency on exercises, and idle duration.
- The agent shall compute a real-time focus score (0–100) and classify engagement states as: *focused*, *distracted*, or *disengaged*.
- Upon detecting disengagement (e.g., idle $>$ 90 seconds, focus score $<$ 25), the agent shall autonomously decide on an intervention: modality switch, micro-break prompt, or difficulty reduction.
- All interventions shall be logged with timestamp, trigger condition, and outcome.

2.5. FR-04: Adaptive Assessment and Feedback

- The Feedback Agent shall generate quizzes adapted to each student's disability profile (short questions, visual multiple-choice, audio options).
- Question difficulty shall adjust dynamically after each response using an Item Response Theory (IRT)-inspired model.
- On incorrect answers, the agent shall provide a targeted explanation using simpler language and an analogy relevant to the student's context.
- Incorrectly answered concepts shall be queued for spaced repetition review.
- The system shall never display "wrong" as a standalone response; feedback shall always include an explanation.

2.6. FR-05: Automated IEP and Progress Reporting

- The IEP Agent shall generate a weekly progress report for each student without requiring teacher input.
- Reports shall include: session statistics, focus trends, topic mastery levels, most effective adaptation strategies, and struggling areas.
- The agent shall produce a structured IEP update draft with measurable goals, recommended accommodations, and evidence from session data.
- Teachers shall review and approve IEP suggestions via a one-click interface.

- Reports shall be exportable as PDF documents suitable for official school use.

2.7. FR-06: Teacher Dashboard

- Teachers shall have access to a dashboard displaying real-time status for all students.
- The dashboard shall surface automated alerts for students requiring urgent attention.
- Teachers shall be able to upload content, review IEP drafts, and adjust student profiles.

3. Technical Requirements

3.1. System Architecture

AdaptLearn is built on a **multi-agent agentic architecture**, where a central Orchestrator Agent coordinates five specialized sub-agents. This design is inspired by the ReAct (Reasoning + Acting) paradigm and implements a continuous observe–plan–act–evaluate loop.

3.1.1. Agent Roles and Responsibilities

Table 2: Agent registry — AdaptLearn multi-agent system

Agent	Responsibility	Color code
Orchestrator	Plans task sequences, delegates to sub-agents, evaluates results	Purple
Profile Agent	Onboards students, builds and updates learner models	Teal
Adaptation Agent	Transforms content to match learner profiles	Purple
Monitor Agent	Tracks engagement signals, triggers interventions	Coral
Feedback Agent	Generates adapted quizzes, provides explanations	Teal
IEP Agent	Aggregates session data, generates progress reports	Coral

3.1.2. Orchestration Pattern

The Orchestrator follows the **ReAct loop**:

1. **Observe** — Receive event (student action, session state, teacher upload).
2. **Reason** — LLM call with full context: determine what is needed and why.
3. **Plan** — Select agent(s) to invoke and construct their input parameters.
4. **Act** — Execute agent calls (sequential or parallel based on dependencies).

5. **Evaluate** — Assess agent outputs; loop back if goal not met.

3.2. Technology Stack

3.2.1. Agent Framework

- `LangGraph` (LangChain) — Primary framework for building the stateful multi-agent graph. LangGraph's `StateGraph` abstraction maps directly to our agent architecture: each node is an agent, edges represent delegation paths, and shared state holds the learner model and session context.
- `LangChain Tools` — Used to define the tool-use capabilities of each agent (web search, TTS, content chunker, database read/write).

3.2.2. Language Models

- `Claude 3.5 Sonnet` (Anthropic) or `GPT-4o` (OpenAI) — Primary reasoning backbone for the Orchestrator and IEP Agent, where long-context reasoning and structured output generation are critical.
- `GPT-4o-mini` / `Claude Haiku` — Used for high-frequency, lower-stakes calls (e.g., difficulty classification, sentence simplification) to optimize latency and cost.

3.2.3. Backend

- `Python 3.11` — Core runtime environment.
- `FastAPI` — REST API layer exposing agent endpoints to the frontend.
- `PostgreSQL` — Persistent storage for learner profiles, session logs, and IEP documents.
- `Redis` — Session state caching for real-time engagement signals.

3.2.4. Frontend

- `React.js` (TypeScript) — Student interface and teacher dashboard.
- `OpenDyslexic` (Google Fonts) — Dyslexia-friendly font rendered dynamically per student profile.
- `Web Speech API` / `ElevenLabs TTS` — Text-to-speech audio generation.

3.2.5. Tooling per Agent

3.3. Data Flow

1. Student accesses the platform via browser. Onboarding chat initiates Profile Agent.
2. Teacher uploads lesson content. Adaptation Agent is triggered via Orchestrator.
3. During a session, the frontend emits behavioral telemetry (scroll, idle, clicks) to the Monitor Agent via WebSocket.
4. Monitor Agent streams engagement scores to Redis; Orchestrator reads and reacts.
5. At session end, results are persisted to PostgreSQL. IEP Agent is scheduled weekly.

Table 3: Tool registry — tools available to each agent

Agent	Tools
Profile Agent	`questionnaire-runner`, `disability-classifier`, `profile-store`
Adaptation Agent	`readability-scorer`, `sentence-simplifier`, `tts-generator`, `content-chunker`, `font-transformer`
Monitor Agent	`scroll-tracker`, `time-on-task`, `engagement-classifier`, `modality-switcher`, `alert-sender`
Feedback Agent	`question-generator`, `difficulty-adjuster`, `explanation-generator`, `spaced-repetition-queue`
IEP Agent	`session-aggregator`, `insight-extractor`, `report-builder`, `pdf-exporter`

6. Teacher reviews dashboard and approves IEP updates.

3.4. Non-Functional Requirements

- **Latency:** Content adaptation response time $<$ 3 seconds for documents up to 5,000 words.
- **Availability:** Target 99.5% uptime during school hours.
- **Privacy:** All student data anonymized for LLM calls. No personally identifiable information (PII) sent to external APIs. GDPR-compliant storage.
- **Accessibility:** WCAG 2.1 AA compliance for the student interface.
- **Scalability:** Architecture supports horizontal scaling via containerized deployment (`Docker` + `Kubernetes`).

3.5. MVP Scope for Qualification Phase

Given the 48-hour constraint of the qualification phase, the MVP focuses on demonstrating core agentic behavior clearly and convincingly. The following features are implemented in the prototype:

Onboarding flow — Profile Agent correctly classifies student profile from chat

Content transformation — Adaptation Agent rewrites and reformats a sample lesson

Attention event simulation — Monitor Agent detects idle state and triggers modality switch

Adaptive quiz — Feedback Agent adjusts difficulty and explains wrong answers

IEP report generation — IEP Agent produces a structured PDF report from session data

4. Why This Project is Truly Agentic

Many education AI products are pipelines or chatbots. AdaptLearn is neither. The distinction matters for the evaluation criteria and is worth stating explicitly.

- **Autonomy:** The Monitor Agent does not wait for instructions. When it detects disengagement, it independently decides to switch modalities — this decision is not hardcoded. It is the result of reasoning over the student's profile, the session history, and the available intervention options.

- **Goal-directedness:** Each agent has a clear objective function. The Adaptation Agent's goal is not "transform text" but "maximize this student's comprehension given their profile." It uses multiple tools and iterations to achieve that goal.

- **Tool use:** Every agent calls external tools (TTS, database queries, readability scorers) that extend its capabilities beyond what the LLM alone can provide.

- **Multi-modal perception:** The system ingests text (lesson content), behavioral telemetry (engagement signals), and structured data (learner profiles) — combining them to make decisions no single-modality system could.

- **Long-horizon planning:** The IEP Agent does not act on a single event. It reasons over weeks of data to identify patterns and make forward-looking recommendations — a hallmark of genuine agentic behavior.

5. Expected Impact

Table 4: Projected impact of AdaptLearn across key stakeholder groups

Stakeholder	Benefit	Metric
Students with LD	Content delivered in their optimal format	Comprehension rate
Teachers	Weekly IEP reports generated automatically	Hours saved per week
School administrators	Scalable support without specialist hiring	Cost per student served
Parents	Transparent progress tracking	Engagement & trust
Policy makers	Evidence base for inclusion initiatives	Coverage of LD population

Research shows that students who receive appropriate accommodations show **21–43% improvement** in academic performance metrics compared to unsupported peers (*Swanson & Hoskyn, Review of Educational Research, 1998*). AdaptLearn makes those accommodations available to every student, every day, at scale — not just to those whose families can afford private specialists.

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